

SYNTHETIC — illustrative, not from any real company. Written for a portfolio RAG project; does not describe any real incident.

Incident Postmortem: Loyalty Points Double-Credit

Incident ID: POS-ERR-3210 | Date: 2026-05-22 | Severity: **SEV-3**

Summary

A deployment of the loyalty accrual service on 2026-05-22 introduced a race condition that, under specific retry-timing conditions, credited points twice for a single transaction. Approximately 1,900 transactions across all stores were affected before the bug was identified and rolled back, roughly 9 hours after deployment.

Root Cause

The accrual service retries a failed write to the points ledger up to 3 times with exponential backoff. The 2026-05-22 deployment changed the retry logic to use a new idempotency key derived from transaction ID and timestamp, intended to prevent duplicate credits on retry. The bug: the timestamp component used millisecond precision from the request, not the original transaction time, so a retried request generated a **different** idempotency key from the original attempt whenever the retry crossed a

millisecond boundary — which, under load, was nearly always.

This meant every retried accrual (not every accrual — only ones that hit a transient failure and were retried) was treated as a new, distinct transaction rather than a duplicate of the original, and credited again. The bug required both a transient failure **and** a retry to manifest, which is why it affected roughly 1,900 of the day's approximately 40,000 transactions rather than all of them.

Detection and Response Timeline

Time	Event
09:14	Deployment completes
14:40	Customer support flags unusual volume of "why do I have more points than expected" tickets
15:05	Engineering confirms double-credit pattern in ledger data
15:20	Deployment rolled back
18:00	Affected accounts identified via ledger diff against pre-deployment baseline

Remediation: Point Reversal

Affected accounts had the duplicate credit reversed via a corrective ledger entry, not a full points-balance reset — this distinction matters because several affected customers had already redeemed some points (including the duplicate portion) between the double-credit and the reversal. Where a customer had already redeemed more points than their correct (non-duplicated) balance would have allowed, the negative balance was written off rather than clawed back from the customer, at a total cost of

approximately \$4,200 across all affected accounts — the engineering team's assessment was that attempting to claw back already-spent points from customers who had no way to know the credit was erroneous would cause disproportionate reputational cost relative to the amount involved.

Interaction with Points Expiry and Returns

A small number of affected accounts had already had the duplicate points partially consumed by a return under Loyalty Program Rules LOY-POL-001 Section 4 (points handling on returns) before the reversal was processed — in those cases, the reversal was applied only to the remaining duplicate balance, since the returned portion had already correctly round-tripped through the normal return-refund mechanism and reversing it a second time would have under-credited the customer relative to their genuine entitlement.

Corrective Actions

1. Idempotency key generation now uses the original transaction timestamp, sourced from the transaction record itself, not the request timestamp of whichever attempt (original or retry) is currently executing.
2. Added a ledger-level uniqueness constraint on (transaction ID, accrual type) as a second line of defense, independent of the idempotency key logic, so a future bug in key generation cannot reproduce this failure mode even if the underlying cause is different next time.
3. Deployment process now requires a synthetic retry-storm load test against a staging ledger before any accrual-service change ships, mirroring the staging-coverage gap identified in the POS outage postmortem (POS-ERR-3021) for a different service.

Related Documents

LOY-POL-001 — Loyalty Program Rules · LOY-POL-002 — Points Expiry Policy · POS-ERR-3021 — POS Outage Postmortem (related staging-coverage gap)